

GO-16: The Adversarial Observer

Statement v0.5 — the revelation reduction and the partition/tie theorem
status: theory netted at LQG scope, R-IND-5-verified, and GOVERNED — opened
2026-08-21 at [exploratory] (v0.1, un-parked from U4 out of
paper/OT-POKER-SURVEY.md); v0.2 same day: Theorems 1–3 proved, harness ALL PASS,
Lean-checked; v0.3 same day: fresh-context R-IND-5 pass complete (2 exposition errors, 3
gaps, 2 overclaims — every finding addressed), novelty flank on record, discrete twin settled
at probe level; v0.4 same day: **governed seals GO-P-2026-090 (ALL PASS 9/9) and
091 (ALL PASS 4/4)** at governed seed 20260822, both in CI’s re-run tier, LEDGER row
added — theorems citable at [predicted]-grade; quote-level citation sweep complete (13
verified / 2 corrected / 1 unverifiable-at-primary of 16, record in the flank file); the
campaign itself remains [exploratory] pending the operational face; v0.5 same day:
Theorem 4 (the binary water-level theorem, theorem-grade — exact channel closed form,
the frontier as a two-line theorem superseding 091’s SLSQP instrument, exchange-lemma
ties), **Theorem 5** (Theorem 3 upgraded to commuting scope), **Theorem 6** (NO
fixed-instrument commitment gap at commuting scope — the open question of v0.3
resolved, with an explicit optimal glance mixture), the commuting-alignment scan refining
Conjecture 1 (the contested phase is OPEN around commuting, not measure-zero), and a
second Lean module (11 lemmas, built clean on Atlas, zero sorry)

Geometric Observation program

2026-08-21 (v0.5)

Abstract

The adversarial observer is the program’s first anti-aligned consumer: a rank- k -budgeted reader of the encoder’s own record, where the record is simultaneously the encoder’s payoff instrument. v0.1 posed the LQG disclosure game and conjectured a partition/tie structure at its optimum. This version proves it, through a reduction the conjecture did not anticipate: leakage depends on the policy (F, Σ_w) only through the **revelation operator** $K = F^\top N^{-1} F$, and the minimal value cost of achieving K is *exactly linear*, $\text{tr}(S(I - K)S^\top)$, attained by **shrink-and-dither** ($F = SK$, $\Sigma_w = SK(I - K)S^\top$). The game is therefore a convex semidefinite program with a bilinear saddle against the reader’s mixed attention operator (Theorem 1). The saddle conditions yield the partition: directions are *revealed*, *blacked out*, or *contested*, contested directions are priced to encoder indifference, and fractional attention forces the spectral tie $\lambda_k = \lambda_{k+1}$ (Theorem 2). At diagonal scope the solution is closed-form: a water level t^* , contested SNRs $\rho_i = t^*/\mu_i$, reader mixing frequencies $\theta_i = s_i^2/(\lambda\mu_i)$, and **tie iff contested** $\neq \emptyset$ (Theorem 3) — each side’s mixture pinned by the other’s indifference, exactly the mixed-equilibrium structure of poker. Two v0.1 items fall: Conjecture 2 (never-dither) is **refuted** — fractional revelation *requires* dither, and the v0.1 probe’s contrary finding is explained by its instances (their optima are near-projections needing $\approx$ no dither; the v0.2 jitter diagnosis was itself wrong and is corrected on the R-IND-5 record) — and Conjecture 4 (commitment gap) collapses *for the whitened-attention game*: with the reader committing to a

mixed attention operator, minimax = maximin; the fixed-instrument Θ -mixture game is left open. The discrete twin (binary alphabet) is settled at probe level: the tie survives exactly, the pricing generalizes, mixing is the carrier of dither, and a fifth class appears that LQG forbids. Verification: numerical harness ALL PASS 9/9 + a diagnosis measurement (`experiments/go16.verify_partition.py`); twin probe ALL PASS 4/4 (`experiments/go16.discrete_twin.py`); six lemmas machine-checked in Lean 4/Mathlib (`lean/ObservationTheory/AdversarialObserver.lean`, built clean on Atlas, zero **sorry**); fresh-context R-IND-5 pass complete — the mathematical core survived intact, and the seven findings (2 exposition errors, 3 gaps, 2 overclaims) are addressed in this version. Novelty flank on record: the composite game is unoccupied; the SDP-ization machinery is imported (Sayin–Akyol–Başar; Tamura), the tie’s optimization ancestor is Overton–Womersley coalescence, and the partition/pricing/water-level theorem is the headline novelty.

1 The object

An observer in this program is a triple $O = (C, G, B)$ inducing a read metric P_C . Everywhere so far the encoder compresses *for* its consumer. Here the encoder’s record A — its observable behavior accumulated over play — is read by two parties: the encoder’s own payoff functional (actions must track the private state to earn value), and an adversary D who reads A through a *budgeted* read operator and converts what it resolves into the encoder’s loss. The encoder’s objective is a two-consumer program with opposite signs. Poker vocabulary maps exactly: the ideal policy S is maximal exploitation; shielding is balance; the reader’s kernel is the bluffing space; reader indifference is the indifference condition that defines mixed equilibria.

2 The LQG disclosure game

Definition 1 (Disclosure game $\mathcal{D}(S, H, k, \lambda)$). Private type $X \sim \mathcal{N}(0, I_n)$. The encoder commits to

$$A = FX + W, \quad W \sim \mathcal{N}(0, \Sigma_w), \quad \Sigma_w \succeq 0 \perp X,$$

with $F \in \mathbb{R}^{m \times n}$, value-ideal $S \in \mathbb{R}^{m \times n}$, and **value loss** $L_{\text{val}} = \|F - S\|_F^2 + \text{tr } \Sigma_w$. The adversary picks a rank- k read Θ (WLOG $\Theta \Theta^\top = I_k$), observes $Y = \Theta A$, and exploits $Z = HX$; the encoder’s **leakage** is the resolved variance $R(\Theta) = \text{tr Cov}(\mathbb{E}[Z | Y])$. With $N = FF^\top + \Sigma_w$ and the encoder committing first (the population/HUD setting),

$$J(F, \Sigma_w) = L_{\text{val}} + \lambda \max_{\Theta} R(\Theta; F, \Sigma_w).$$

Throughout, $M := H^\top H$ (the exploit Gram) and $\Sigma_S := S^\top S$ (the value Gram). Regularity: policies with $N \succ 0$; singular N is handled by the εI limit, which changes no infimum.

3 Elementary structure (v0.1, kept)

Proposition 1 (Kernel-freeness). *For fixed Θ with read projection $P = \Theta^\top \Theta$, every minimizer satisfies $(I - P)F^* = (I - P)S$ and $\Sigma_w^* = P\Sigma_w^*P$: the free-exploitation subspace is exactly $\ker \Theta$ (dimension $m - k$), carrying the exploitative policy in full, with no dither.*

Proposition 2 (The budgeted reader's value). $\max_{\Theta} R = \sum_{i=1}^k \lambda_i(G) = \|G\|_{(k)}$ with $G = N^{-1/2} F H^{\top} H F^{\top} N^{-1/2}$, attained on the top- k eigenspace of G ; $\|G\|_{(k)}$ is constant in k beyond $\text{rank}(H F^{\top})$ (rank saturation).

Proofs as in v0.1 (the substitution $U = N^{1/2} \Theta^{\top}$ and Ky Fan's maximum principle).

4 The revelation reduction

Definition 2 (Revelation operator). $K = K(F, \Sigma_w) := F^{\top} N^{-1} F \in \mathbb{R}^{n \times n}$, the posterior precision gain about X from the full record: $\text{Cov}(X | A) = I - K$. Always $0 \preceq K \preceq I$.

Theorem 1 (Revelation reduction). (i) (Sufficiency of K .) *The leakage spectrum is a function of K alone: the nonzero eigenvalues of G and of $M^{1/2} K M^{1/2}$ coincide, so $\|G\|_{(k)} = \|M^{1/2} K M^{1/2}\|_{(k)}$.*

(ii) (Linear cost.) *For $m \geq n$ and S of full column rank, the minimal value cost to achieve revelation K is*

$$C(K) = \text{tr}(S(I - K)S^{\top}),$$

*attained by the **shrink-and-dither policy** $F^* = SK$, $\Sigma_w^* = SK(I - K)S^{\top}$ (so $N^* = SKS^{\top}$). (K with nontrivial kernel: by the ε -limit; $m < n$: the constraint $\text{rank } K \leq m$ appears and is left open.)*

(iii) (Convexity and the saddle.) *The game reduces to the convex semidefinite program*

$$\min_{0 \preceq K \preceq I} \text{tr}(\Sigma_S(I - K)) + \lambda \|M^{1/2} K M^{1/2}\|_{(k)} = \min_{0 \preceq K \preceq I} \max_{W \in \mathcal{W}_k} \mathcal{L}(K, W),$$

$\mathcal{L}(K, W) := \text{tr}(\Sigma_S(I - K)) + \lambda \text{tr}(W M^{1/2} K M^{1/2})$, $\mathcal{W}_k := \{0 \preceq W \preceq I, \text{tr } W = k\}$; both feasible sets are compact convex and $\mathcal{L}$ is bilinear, so a saddle point (K^*, W^*) exists and minimax equals maximin.

Proof. (i) With $B = N^{-1/2} F$ we have $G = B M B^{\top}$ and $K = B^{\top} B$; the nonzero spectra of $B M B^{\top}$ and $M^{1/2} B^{\top} B M^{1/2}$ coincide (XY vs. YX). Top- k sums of equal nonzero spectra of PSD matrices agree.

(ii) *Lower bound.* Write $F = N^{1/2} B$, $P = N^{1/2} \succ 0$, $Z = S B^{\top}$, $Q = \text{sym } Z = \frac{1}{2}(Z + Z^{\top})$. Then

$$\text{cost} = \|N^{1/2} B - S\|_F^2 + \text{tr } N - \text{tr}(F F^{\top}) = \text{tr}(S S^{\top}) + \text{tr } P^2 - 2 \text{tr}(P Q),$$

using $\text{tr}(B^{\top} N B) = \text{tr}(N B B^{\top})$ (which cancels) and $\text{tr}(P Z) = \text{tr}(P Q)$ (symmetric $\times$ antisymmetric is traceless). Since $\text{tr}((P - Q)^2) \geq 0$ for symmetric $P - Q$, $\text{tr } P^2 - 2 \text{tr}(P Q) \geq -\text{tr } Q^2$. Finally

$$\text{tr } Q^2 = \frac{1}{2} [\text{tr}(Z^2) + \text{tr}(Z Z^{\top})] \leq \text{tr}(Z Z^{\top}) = \text{tr}(S B^{\top} B S^{\top}) = \text{tr}(S K S^{\top}),$$

because $\text{tr}(Z Z^{\top}) - \text{tr}(Z^2) = \frac{1}{2} \|Z - Z^{\top}\|_F^2 \geq 0$. The converse thus uses *two* inequalities, both machine-checked: $\text{tr}(X X^{\top}) \geq 0$ (Lean: `trace_mul_transpose_self_nonneg`, applied to symmetric $X = P - Q$ and to $Z - Z^{\top}$) and $\text{tr}(Z^2) \leq \text{tr}(Z Z^{\top})$ (Lean: `trace_sq_le_trace_mul_transpose`). Hence $\text{cost} \geq \text{tr}(S S^{\top}) - \text{tr}(S K S^{\top}) = \text{tr}(S(I - K)S^{\top})$.

Achievability. For $F^* = SK$, $\Sigma_w^* = SK(I - K)S^{\top}$ (PSD since $\text{spec } K \subset [0, 1]$): $N^* = SK^2 S^{\top} + SK(I - K)S^{\top} = SKS^{\top}$ by the ring identity $K^2 + K(I - K) = K$ (Lean:

revelation_key, revelation_variance — corrected from v0.2’s erroneous citation of the telescope identity, R-IND-5 finding 11); the cost telescopes to $\text{tr}(S(I - K)S^\top)$ by $(I - K)^2 + K(I - K) = I - K$ (Lean: *shrink_dither_key, shrink_dither_cost*). K -verification (display corrected from v0.2, R-IND-5 finding 4): with $S = U\Sigma V^\top$ (thin SVD; V is $n \times n$ orthogonal by full column rank, U has orthonormal columns) and $K \succ 0$,

$$N^* = U(\Sigma V^\top K V \Sigma) U^\top, \quad (N^*)^+ = U(\Sigma V^\top K V \Sigma)^{-1} U^\top,$$

(the four Moore–Penrose conditions hold since U has orthonormal columns), whence

$$F^{*\top} (N^*)^+ F^* = K V \Sigma (\Sigma V^\top K V \Sigma)^{-1} \Sigma V^\top K = K V (V^\top K V)^{-1} V^\top K = K,$$

using $\Sigma(B\Sigma)^{-1}\Sigma = B^{-1}$ with $B = V^\top K V$ and $V V^\top = I$. Singular K : by the ε -limit (asserted-standard prose; the harness verifies exact attainment for singular and projection K directly, so the identity in fact holds beyond the stated scope — R-IND-5 finding 4).

(iii) $\|\cdot\|_{(k)}$ is the support function of $\mathcal{W}_k$ (Ky Fan), giving the inner max; linear-plus-support-function is convex; Sion/von Neumann gives the saddle. $\square$

Corollary 1 (Deception requires dither; v0.1 Conjecture 2 **refuted**). *A noiseless record ($\Sigma_w = 0$, F full row rank) has $K = F^\top (F F^\top)^{-1} F$ idempotent: revelation is a projection, all-or-nothing per direction — rescaling a noiseless record shields nothing (the reader renormalizes). Fractional revelation — which the optimum generically demands (Theorem 3) — is achievable only with $\Sigma_w \neq 0$. Optimal deception is shrink-and-dither: randomization is intrinsic, not a discrete-support artifact. **v0.1’s Conjecture 2 asserted the opposite; it is refuted by this corollary** — the diagonal instances demand dither trace 0.19 at the optimum. **The diagnosis of the v0.1 probe finding (C4, “never dithers”), however, has itself been corrected twice, and both corrections stay on the record.** v0.2 attributed C4 to an instrument artifact (εI jitter supplying free noise). The R-IND-5 pass refuted that mechanism quantitatively (the jitter route books blackout-level cost at mid-range revelation — finding 8), and the V10 measurement settles it: at the v0.1 probe’s own (λ, k) cells, the exact optima are near-projections (fractional mass $\leq 10^{-13}$, required dither $\leq 2 \times 10^{-12}$ at three of four cells; 0.11 and 0.027 at the fourth). The v0.1 instances needed $\approx$ no dither; they were unrepresentative, not mismeasured. The same measurement yields an unlooked-for structural observation: the contested phase appears to concentrate on commuting-aligned (Σ_S, M) pairs (the diagonal family and its rotations), with generic non-commuting instances optimizing at near-projections — folded into Conjecture 1. The v0.1 poker-facing reading still inverts as v0.2 said: mixing in discrete games is the shadow of the dither requirement, now confirmed by the twin probe (§7).*

Corollary 2 (No commitment gap at LQG scope; v0.1 Conjecture 4 resolved). *By Theorem 1(iii), $\text{minimax} = \text{maximin}$ for the reduced game in which the reader’s strategy is the whitened attention operator W — equivalently, a reader who may condition its read on the record’s law N (the population/HUD setting: the pool reads the frequencies of the record actually played). In that game, encoder commitment is worthless. **Scope correction (R-IND-5 finding 9):** this does not establish the claim for the concrete game in which the reader commits to a mixture over fixed instruments Θ without knowledge of N : the map from Θ to attention, $P_U = N^{1/2} \Theta^\top (\Theta N \Theta^\top)^{-1} \Theta N^{1/2}$, depends on the encoder’s N , so a committed Θ -mixture cannot enforce a fixed W^* across encoder deviations. Whether a commitment gap*

exists in the fixed-instrument game is **open**. What stands: the reader must mix (projections — the extreme points of $\mathcal{W}_k$ — announced first are exploitable), and the poker structure is recovered, not assumed, in the N -conditioned reading.

5 The partition/tie theorem

Theorem 2 (Partition and ties, general scope). *Let (K^*, W^*) be any saddle point of $\mathcal{L}$, and set $G^* = M^{1/2}K^*M^{1/2}$, $\Delta^* = \lambda M^{1/2}W^*M^{1/2} - \Sigma_S$. Then:*

- (i) (Pricing.) K^* minimizes the linear functional $\text{tr}(K\Delta^*)$ over $0 \preceq K \preceq I$: on the negative eigenspace of Δ^* , $K^* = I$ (**revealed** — the conceded directions, where even full attention is priced below value, together with the submerged directions the reader ignores); on the positive eigenspace, $K^* = 0$ (**blacked out**); every eigenvector of K^* with fractional eigenvalue lies in $\ker \Delta^*$ (**contested**): on contested directions the reader’s mixed attention prices revelation to exact encoder indifference, $\lambda M^{1/2}W^*M^{1/2} = \Sigma_S$.
- (ii) (Attention.) W^* maximizes $\text{tr}(WG^*)$ over $\mathcal{W}_k$: $W^* = P_{>} + Z$ with $P_{>}$ the projector onto eigenvalues $> \lambda_k(G^*)$ and Z supported on the $\lambda_k(G^*)$ -eigenspace.
- (iii) (Tie.) If W^* has an eigenvalue in $(0, 1)$ — the reader mixes — then $\lambda_k(G^*) = \lambda_{k+1}(G^*)$.

Proof. (i) and (ii) are the saddle conditions; both subproblems are linear over spectral polytopes, so optimizers are as described (for (ii), Ky Fan’s maximum principle with its equality cases). (iii): suppose $\lambda_k > \lambda_{k+1}$. The eigenvalue group of value λ_k then ends at position k ; maximizers put weight 1 on all eigendirections with value $> \lambda_k$ (count $k - r$, where r is the multiplicity of λ_k) and the remaining budget r exactly fills the group — forcing $W^* = P_{\geq}$, a projection. Contrapositive gives the claim. $\square$

Theorem 3 (Diagonal closed form: the attention water level). *Let $\Sigma_S = \text{diag}(s_1^2, \dots, s_n^2)$, $M = \text{diag}(\mu_1, \dots, \mu_n)$, under genericity: (G1) μ_i distinct, positive; (G2) $\lambda\mu_i \neq s_i^2$ for all i . Then the optimum of the reduced program is diagonal, $K^* = \text{diag}(\rho_i)$, with a **water level** $t^* \geq 0$ at a breakpoint $t^* = \mu_j$, and the solution falls in exactly one of two regimes: the **fractional regime** — condition (G3): the attention-budget equation resolves with strictly interior slack — in which the contested set is nonempty, or the **integral regime**, in which every $\theta_i \in \{0, 1\}$ and the contested set is empty. (Scoping corrected from v0.2, whose (G3) hypothesis made the empty-contested branch vacuous — R-IND-5 finding 7: G3 forces contested $\neq \emptyset$; the two regimes are now stated as the exhaustive alternative.) With $g_i := \mu_i\rho_i$:*

- **Conceded** ($\lambda\mu_i < s_i^2$, $\mu_i > t^*$): $\rho_i = 1$, $g_i = \mu_i$, attention $\theta_i = 1$. (The lazy regime: shielding costs more than the priced leak.)
- **Contested** ($\lambda\mu_i > s_i^2$, $\mu_i > t^*$): $\rho_i = t^*/\mu_i \in (0, 1)$, $g_i = t^*$ exactly, and $\theta_i = s_i^2/(\lambda\mu_i) \in (0, 1)$: the reader’s mixing frequency equals the ratio of shielding cost to priced attention — pinned by encoder indifference.
- **Level** ($\mu_j = t^*$): $\rho_j = 1$, absorbing the attention slack $\theta_j = k - \#\{\text{conceded}\} - \sum_{\text{contested}} s_i^2/(\lambda\mu_i)$, capped by $\min(1, s_j^2/(\lambda\mu_j))$; $\theta_j \in (0, 1)$ in the fractional regime, $\theta_j \in \{0, 1\}$ in the integral one.

- **Submerged** ($\mu_i < t^*$): $\rho_i = 1, g_i = \mu_i, \theta_i = 0$: fully revealed, unread — free exploitation by budget exhaustion rather than by structural invisibility.

Moreover the **tie** $\lambda_k(G^*) = \lambda_{k+1}(G^*) = t^*$ holds iff the contested set is nonempty — equivalently, iff the solution is in the fractional regime; in the integral regime (e.g. $\mu = (4, 2.5, 1.8)$, $s^2 = (5, 5, 5)$, $\lambda = 1$, $k = 2$: all conceded, $\theta = (1, 1, 0)$, spectrum strict — the R-IND-5 verifier’s witness) the top- k spectrum is strict under (G1).

Proof. Diagonality (symmetrization). The reduced program is convex (Theorem 1(iii)) and invariant under $K \mapsto DKD$ for every sign matrix $D = \text{diag}(\pm 1)$ (both Σ_S and M commute with D ; the spectrum of $M^{1/2}DKDM^{1/2} = DM^{1/2}KM^{1/2}D$ is unchanged). Averaging K over the 2^n sign conjugations preserves feasibility, does not increase the convex objective, and returns $\text{diag}(K)$; a diagonal optimum exists.

Per-coordinate cost. At diagonal K , coordinate i carries SNR ρ_i at cost $s_i^2(1 - \rho_i)$ — the scalar case of Theorem 1(ii), by the exact completion of squares $(f - s)^2 + f^2(1 - \rho)/\rho = s^2(1 - \rho) + (f - \rho s)^2/\rho$ (Lean: **scalar_shielding_identity**, **scalar_shielding_cost_bound**), minimized at the shrink-and-dither point $f = \rho s$, $w = \rho(1 - \rho)s^2$.

Water level. With $\phi_k(g) = \min_t [kt + \sum_i (g_i - t)_+]$, the program is jointly convex in (ρ, t) and separable at fixed t : coordinate i ’s objective $s_i^2(1 - \rho_i) + \lambda(\mu_i \rho_i - t)_+$ has constant slopes $-s_i^2 + \lambda\mu_i > 0$ above the kink and $-s_i^2 < 0$ below (G2), so its minimum is at the kink $\rho_i = \min(1, t/\mu_i)$ when $\lambda\mu_i > s_i^2$, and at $\rho_i = 1$ otherwise: linear-versus-linear forces bang-bang-to-the-kink — this is where the tie mechanism lives. The value $J(t)$ is piecewise linear with breakpoints $\{\mu_i\}$, so its minimum sits at a breakpoint $t^* = \mu_j$ (generically unique); the subgradient condition $0 \in \partial J(t^*)$ is the attention budget $\sum_i \theta_i = k$ with the stated θ , and the caps are exactly the encoder’s first-order conditions at $\rho_i = 1$ boundaries.

Tie. If contested = $\{i_1, \dots, i_c\}$, $c \geq 1$: the spectrum of G^* has the conceded values above t^* (count a), then the tied group $\{t^*\}$ of multiplicity $c + 1$ (contested plus level), then the submerged values. The budget gives $k = a + \sum_{\text{contested}} \theta_i + \theta_j$ with every θ in the group fractional, so $a < k < a + c + 1$, hence both λ_k and λ_{k+1} lie in the tied group: $\lambda_k = \lambda_{k+1} = t^*$. If contested = $\emptyset$ the spectrum is $\{\mu_i\}$, distinct by (G1): strict. $\square$

Remark 1 (The poker reading). Contested directions are the mixing support: there, and only there, (a) the reader mixes (fractional θ = randomized attention — bluff-catching frequencies), (b) the encoder part-shields (fractional ρ = balanced ranges), (c) each side’s frequency is pinned by the *other’s* indifference ($\theta_i = s_i^2/(\lambda\mu_i)$; $g_i = t^*$), and (d) the spectrum ties — indifference holds on the support only, exactly as in poker. Conceded directions are “too cheap to balance” (the lazy regime the probe found); submerged directions are exploitation for free by budget exhaustion; $\ker \Theta$ (fixed reads) was v0.1’s structural free-exploitation and survives as the W -null space.

6 Verification

Numerical harness (experiments/go16_verify_partition.py, seed 20260821, deterministic; development netting run — the governed seal is owed): **ALL PASS 9/9**. Achievability algebra exact (cost error 0.0, K -error 1.3×10^{-11} , spectrum equality 5.9×10^{-11}); cost lower bound: 0 violations in 2000 random policies; noiseless-record idempotency 2.9×10^{-14} (dither

necessity); hand-derived 6-coordinate instance reproduced to machine precision ($t^* = 1.0$, $J^* = 4.05277\dots$, tie gap 0.0, $\theta = (0.125, 1, 0.2\bar{2}, 0.6527\dots, 0, 0)$, $\sum \theta = k = 2$, all four classes populated); direct full-matrix minimization of the original game lands on the closed form (4.0565 vs. 4.0528, optimizer above the true minimum as it must be); saddle survives 500 random perturbations per side with zero violations, dither trace at optimum 0.19; the exact $m = 2$ example gives $J^* = 7/4$, $t^* = 1$, $\theta = (1/4, 3/4)$ exactly. V9 (de-vacuated per R-IND-5 finding 10b): a *rotated* diagonal instance with known fractional optimum — the generic SDP solver recovers the fractional revelation spectrum in a non-diagonal basis (spectrum error 6.6×10^{-3} vs. truth, tie gap 7×10^{-4} relative), plus the original random general instance (SDP vs. direct J -match). V10 (measurement, not a gate): the v0.1-instance revelation spectra behind the corrected C4 diagnosis (Corollary 1). Gate-quality notes from R-IND-5 finding 10 are carried in-code: V6 is circular with the theorem by construction (V5 direct-solve and V7’s 1000-perturbation saddle check are the non-circular support), and the 10^{-12} jitter in the harness’s own N is commented at its definition.

Lean (lean/ObservationTheory/AdversarialObserver.lean; Lean v4.32.2, Mathlib v4.32.2, built clean 2026-08-21 on Atlas, 8663 jobs, zero errors, zero sorry; rebuilt clean after the v0.3 additions): six lemmas — `shrink_dither_key` / `shrink_dither_cost` (the cost telescope), `revelation_key` / `revelation_variance` (the record-variance identity $N^* = SKS^\top$, added in v0.3 to close R-IND-5 finding 11’s mis-attribution), `trace_mul_transpose_self_nonneg` and `trace_sq_le_trace_mul_transpose` (the converse’s two inequalities), `scalar_shielding_identity` / `_cost_bound` (the per-coordinate completion of squares). *Unformalized remainder, named (extended per R-IND-5 finding 11)*: the Gaussian-conditioning derivations ($R(\Theta)$ and $\text{Cov}(X | A) = I - K$), the AB/BA nonzero-spectrum coincidence of Theorem 1(i), Ky Fan’s maximum principle, saddle existence (Sion), the sign-symmetrization diagonality argument of Theorem 3(a), the ε -limit assertions, the SVD computation in achievability’s K -verification, and the KKT bookkeeping of Theorems 2–3; these are standard-cited and harness-netted.

Lean, second module (lean/ObservationTheory/AdversarialObserverDiscrete.lean; same toolchain, rebuilt clean 2026-08-21 on Atlas, 8664 jobs, zero errors, zero sorry): eleven lemmas for §8 — `evan_closed_form` / `rho_closed_form` (the exact channel algebra), `frontier_bound` / `frontier_attained` / `symmetric_rho` (the frontier theorem), `sqrt_midpoint_concave` (frontier-cost convexity), `shield_foc_identity` / `_bound` (the fifth-class FOC), `sign_flip_average` (the symmetrization step of Theorems 3/5), `attention_exchange` / `fractional_pair_tie` (the reader-side exchange lemma and the discrete tie), `junk_domination` (Theorem 6’s row reduction). *Unformalized remainder, named*: the probabilistic EVar derivation, the water-level assembly and integer- k counting, joint diagonalization and convex averaging (Theorem 5), and Theorem 6’s assembly — standard-cited and netted by S1–S3 plus the governed 090/091 harnesses.

Theory-extensions harness (experiments/go16_theory_extensions.py, seed 20260821, dev-disclosed, ALL PASS 3/3, results/go16-theory-extensions.json): S3 exact-identity gates (ρ closed form 3.3×10^{-16} , frontier bound violation 1.3×10^{-15} , symmetric attainment 8.2×10^{-15} , FOC identity 4.0×10^{-15}); S1 the commuting-alignment scan; S2 the fixed-instrument gap measurements. A governed seal for this harness is owed if its results are ever promoted beyond the theorem-netting role (the theorems themselves are carried by proof + Lean).

Correction ledger for the dither question (all steps kept at equal prominence):

v0.1 conjectured never-dither from probe C4; v0.2 refuted the conjecture (Corollary 1) but misdiagnosed C4 as a jitter artifact; the R-IND-5 pass refuted the diagnosis; V10 measured the truth (near-projection optima at the v0.1 cells). The full narrative is in Corollary 1.

R-IND-5 record (fresh-context adversarial verification, 2026-08-21). Independent re-derivation of all twelve components; verdict: *2 errors, 3 gaps, 2 overclaims out of 12 checks — mathematical core intact* (every theorem confirmed; both hand instances reproduced exactly under the verifier’s own solvers; the hardest-flagged SVD step verified fully and found to hold beyond its stated scope). Dispositions in this version: garbled SVD display rewritten (finding 4); Theorem 3’s (G3)/iff vacuity re-scoped as the two-regime alternative (finding 7); the C4 jitter diagnosis withdrawn and replaced by the V10 measurement (finding 8); Corollary 2 restated for the whitened-attention game with the fixed-instrument game left open (finding 9); V9’s tie gate de-vacuated and the harness circularity notes added (finding 10); the Lean mis-attribution closed by **revelation_variance** and the unformalized-remainder list extended (finding 11); abstract phrasing aligned with the corrected scopes (finding 12).

7 The discrete twin (probe-level)

`experiments/go16_discrete_twin.py` (deterministic; exploratory, disclosed; **ALL PASS 4/4**, `results/go16-discrete-twin.json`): the binary disclosure game — n spots, private bits, randomized bet/check policies (q_0, q_1) , value loss $s_i^2 \Pr[A_i \neq X_i]$, a reader watching k of n spots, leakage μ_i times the resolved variance ratio ρ_i . Findings, each the predicted deformation of the LQG theorem or sharper:

- **The binary shielding frontier is the symmetric channel exactly:** $e(\rho) = (1 - \sqrt{\rho})/2$ (SLSQP-refined to 10^{-12} ; an apparent asymmetric advantage was a binning artifact, on record). Convex, not linear: *the LQG linear cost is a Gaussian privilege* (shape deviation up to 0.125).
- **The tie survives exactly** (it is reader-side, curvature-independent): the transplanted 6-spot instance gives a five-way tie at $t^* = 0.36754$ with spread 0.0, attention budget exact to 4×10^{-16} , two-sided saddle clean, and an independent grid solver agreeing to 1.1×10^{-3} . The pricing generalizes to $\theta_i = -c'_i(\rho_i)/(\lambda\mu_i)$, closed form $\theta_i = s_i^2/(4\lambda\sqrt{\mu_i t^*})$.
- **A fifth class appears that LQG forbids:** *interior-contested-above* — convex cost admits an interior first-order balance strictly above the water with $\theta = 1$ and no tie (“partial balancing without indifference”), exhibited exactly at the hand prediction ($\rho_1 = 0.5625$, g -gap 1.25, $J^* = 3.75$). In LQG, linear cost forces bang-bang to the tie; discretely, a spot can be partially balanced while the opponent still always watches it.
- **Mixing is the discrete carrier of dither:** deterministic policies give $\rho \in \{0, 1\}$ only (the exact twin of idempotent noiseless revelation); every shielded spot at the optimum plays strictly randomized bluff frequencies (reported per spot). And *the first bluff is cheap*: the frontier slope at full revelation is $s^2/4$ vs. LQG’s s^2 , so the conceded class shrinks — in the transplanted instance even the LQG-conceded spot mixes.

v0.2's Conjecture 2 (the discrete twin) is thereby settled at probe level, with the class structure *refined* rather than merely confirmed; the theorem-grade write-up follows in §8.

8 The binary theorem, commuting scope, and the fixed-instrument game (v0.5)

All essential algebra of this section is machine-checked in `lean/ObservationTheory/AdversarialObserverDisc` (cited per step); the exact identities are additionally netted by `experiments/go16_theory_extensions.py` (S3, all gates at 10^{-13} – 10^{-15}).

Theorem 4 (The binary water-level theorem). *In the binary disclosure game of §7 with per-spot policy (q_0, q_1) : (i) (Exact channel algebra.) The resolved-variance ratio has the closed form*

$$\rho(q_0, q_1) = \frac{(q_1 - q_0)^2}{(q_0 + q_1)(2 - q_0 - q_1)} \quad (\text{Lean: } \texttt{rho_closed_form}),$$

on the nondegenerate domain, with the convention $\rho = 0$ at the constant channels. (ii) (The frontier, as a theorem.) Since $u(2 - u) \leq 1$ for $u = q_0 + q_1$, every policy satisfies $\text{err} \geq (1 - \sqrt{\rho})/2$, with equality exactly on the symmetric family $q_0 + q_1 = 1$ (Lean: `frontier\_bound`, `frontier\_attained`, `symmetric\_rho`); deterministic policies have $\rho \in \{0, 1\}$. The 091 pilot's SLSQP refinement is hereby superseded by proof. The frontier cost $c(\rho) = s^2(1 - \sqrt{\rho})/2$ is convex (midpoint concavity of $\sqrt{\cdot}$, Lean: `sqrt\_midpoint\_concave`, plus continuity). (iii) (Saddle and classes.) The reduced game $\sum_i c_i(\rho_i) + \lambda \phi_k(\mu \circ \rho)$ is convex; a saddle against the attention polytope $\{\theta \in [0, 1]^n, \sum \theta = k\}$ exists (Sion). At a saddle, each coordinate's response to θ_i is governed by the exact completion of squares $s^2(1 - x)/2 + cx^2 = s^2/2 - s^4/(16c) + c(x - s^2/(4c))^2$ ($x = \sqrt{\rho}$, $c = \lambda \theta_i \mu_i$; Lean: `shield\_foc\_identity`, `_bound`), giving the five classes measured by 091: conceded ($s_i^2 \geq 4\lambda \mu_i$, $\theta_i = 1$), interior-contested-above (interior FOC, $\theta_i = 1$), tied (fractional $\theta_i = s_i^2/(4\lambda \sqrt{\mu_i t^})$), level, submerged; the water level solves the attention-budget equation $D(t^*) = k$. (iv) (Ties.) Any optimal attention vector satisfies the exchange property — it cannot be interior at a strictly leakier coordinate while positive at a less leaky one (Lean: `attention\_exchange`) — so any two fractional coordinates carry equal g (Lean: `fractional\_pair\_tie`); since k is an integer, a lone fractional coordinate is impossible ($\sum \theta$ would not be an integer), so fractional attention forces a genuine tie $\lambda_k = \lambda_{k+1}$. The tie is reader-side and alphabet-independent, exactly as §7 measured.*

Theorem 5 (Commuting scope). *If $[\Sigma_S, M] = 0$, Theorem 3 holds verbatim in the joint eigenbasis, with (s_i^2, μ_i) the jointly paired eigenvalues: conjugating K by joint-eigenbasis sign flips leaves both terms invariant (the entrywise step is Lean: `sign\_flip\_average`), so averaging over the 2^n flips yields a jointly-diagonal optimum by convexity. The governed face is 090's V9 (the rotated instance — a commuting pair — whose fractional spectrum and tie the generic SDP solver recovers).*

Theorem 6 (No fixed-instrument commitment gap at commuting scope). *In the commuting (WLOG diagonal) game, let (ρ^*, θ^*) be the saddle of Theorem 3 (or its binary twin) with value V^* . Consider the fixed-instrument reader who commits, before seeing anything, to a random single-coordinate glance drawn with marginals θ^* (k glances in expectation; for $k = 1$ this is exactly the fixed rank-one-instrument game over all mixtures of reads). Then the*

encoder’s infimum over ALL policies (F, Σ_w) — including non-diagonal ones — equals V^* : **the v0.3 open question is resolved — there is no commitment gap at commuting scope, and the optimal committed mixture is explicit** (eigenbasis glances at the saddle attention).

Proof. $(\leq)$ is the minimax inequality: the saddle policy is held to exactly V^* by every mixture (at the saddle, every glance resolves its coordinate’s saddle leakage). $(\geq)$: against axis glances the objective *separates across record rows* — the glance at row i sees only $N_{ii} = \|f_i\|^2 + w_{ii}$ (f_i the i -th row of F), the value loss and $\text{tr } \Sigma_w$ are row/diagonal sums, and off-diagonal Σ_w entries appear nowhere. Within a row, moving cross-signal mass into dither ($f_{ij} \mapsto 0$, $w_{ii} \mapsto w_{ii} + f_{ij}^2$, $j \neq i$) leaves the value cost and the read variance unchanged and weakly lowers the resolved numerator (Lean: `junk_domination`), so scalar rows are optimal; the scalar row problem reduces to the shrink-and-dither cost (Lean: `scalar_shielding_identity`, `_cost_bound`), giving row value $\geq \min_\rho [s_i^2(1-\rho) + \lambda\theta_i^* \mu_i \rho]$; summing and using the saddle property of (ρ^*, θ^*) gives $\geq V^*$. Netted: S2 of `go16_theory_extensions.py` measures the gap at 1.4×10^{-12} ($m = 2$ exact instance, full-matrix encoder search over mixtures including three-atom non-axis families) and -1.4×10^{-10} (the governed 6-coordinate instance). *Open residue*: the rank- k *joint-subset* instrument variant (a committed mixture of k -subsets read jointly) is upper- bounded by V^* but its converse needs a subadditivity argument not given here; and the non-commuting fixed-instrument game remains open. $\square$

The commuting-alignment scan (S1, exploratory). The V10 hypothesis is refined by measurement: along random symmetric perturbations of a commuting pair, the contested (fractional revelation) mass stays *high* — 0.49–0.77 for commutator norms up to ≈ 2 — and collapses (≤ 0.02) only under strong misalignment (norms ≈ 2.3 –4.6). **The contested phase is an open region around the commuting stratum, not a measure-zero coincidence**; what vanishes is contested mass in the strongly misaligned regime. Conjecture 1 is updated accordingly: the refined question is the alignment functional governing the phase boundary.

9 What remains open

Conjecture 1 (General-scope converse — refined by V10). *At general (non-diagonal) scope, for generic (Σ_S, M, λ) : $\text{contested} \neq \emptyset \Rightarrow W^*$ fractional (hence the tie, by Theorem 2(iii)); the missing step is ruling out a projection W^* satisfying $\lambda M^{1/2} W^* M^{1/2} = \Sigma_S$ on a nontrivial subspace. The V10 measurement suggested concentration on commuting-aligned pairs; the S1 scan (§8) refines this: the contested phase is an **open region** around the commuting stratum (mass 0.49–0.77 at commutator norms up to ≈ 2), collapsing only under strong misalignment. The refined question: the alignment functional governing the phase boundary — what quantity, computable from $(\Sigma_S, M, \lambda, k)$, separates the contested-rich neighborhood of commuting from the near-projection regime V10 measured at generic far-from-commuting instances?*

Discrete twin (v0.2 Conjecture 2): theorem-grade (Theorem 4; governed face 091). **Signed-weight reduction (v0.1 Conjecture 3), resolved in form:** revelation coordinates convexify the signed-weight two-consumer program; the attention water level with the class partition replaces the two-water-level geometry. **Commitment gap (v0.1**

Conjecture 4): no gap for the whitened-attention game (Corollary 2); **no gap for the fixed-instrument glance-mixture game at commuting scope** (Theorem 6 — the v0.3 open question, resolved with an explicit optimal mixture). Still open: the rank- k joint-subset instrument converse, and the non-commuting fixed-instrument game.

10 Novelty position and campaign gates

Full flank record: `paper/G016-NOVELTY-FLANK.md` (one agent pass, 2026-08-21; 16 must-cites; quote-level pre-seal sweep owed). Posture adopted per its verdict: **the composite game is unoccupied; the machinery is presented as imported**. Specifically: parametrizing Gaussian Stackelberg disclosure by posterior covariance and SDP-izing it is the Sayin–Akyol–Başar / Tamura program (must-cite); the Ky Fan/fantope operator theory is classical (Fan; Overton–Womersley; Warmuth–Kuzmin), and the eigenvalue tie’s optimization-theoretic ancestor is *Overton–Womersley coalescence*; idempotent noiseless disclosure and no-commitment-gap phenomena have published Gaussian instances (Sayin–Başar deception; arXiv:2005.05743) and zero-sum folklore (Conitzer) behind them. **What survived every search angle**, phrased per the quote-level sweep’s essential nuance (the Sayin–Başar SDP objective is *already* linear in the posterior covariance — $\min \text{Tr}\{SV\}$ — so linearity-in-a-covariance-variable is not claimable): the **identification of the minimal value cost of revelation K with $\text{tr}(S(I - K)S^\top)$ inside the record-as-instrument game, with its explicit shrink-and-dither attainment**; and above all the **partition / indifference-pricing / attention-water-level / tie-iff-contested theorem** — the headline. The recent close call arXiv:2602.19292 (Munif–Varma–Lasaulce, IFAC WC 2026: same S -mismatch instrument, spectral disclosure, noiseless all-or-nothing) was verified at quote level against its full text with all four absences confirmed (no budgeted reader, no leakage pricing, no tie/partition theorem, no revelation-cost identification) — cite and differentiate explicitly. The poker/Blotto/security-game equalization analogies are published folklore and are attributed as such (with the sweep’s two corrections: Başar 1983 is the *scalar* jamming saddle — water-filling language belongs to its vector extensions — and Miao–Wu–Young solve by SDP with water-filling only in special cases).

Gates: theory gate **CLEARED, VERIFIED, and GOVERNED**: seals GO-P-2026-090 (partition/tie C3 net, **ALL PASS 9/9**) and GO-P-2026-091 (discrete twin net, **ALL PASS 4/4**) executed at governed seed 20260822 with fresh randomness, sealed pre-run at commit 3a74638 with every prior run disclosed as a pilot; both harnesses in CI’s re-run tier; LEDGER row added — Theorems 1–3 and the twin results are citable at **[predicted]**-grade. The quote-level citation sweep is complete (13/2/1 of 16; the two corrections and the linear-cost phrasing nuance are folded into this version and the flank file). v0.5 discharges three of the four campaign items: the binary write-up is theorem-grade (Theorem 4), the fixed-instrument gap is resolved at commuting scope (Theorem 6), and Conjecture 1 is sharpened to a phase-boundary question by the S1 scan. Next in the campaign: the alignment functional (Conjecture 1); the joint-subset and non-commuting fixed-instrument residues; then the operational face (query-only read-operator recovery of a static pool bot, with exploitation constructed in the kernel and submerged classes and measured leakage) — the campaign node stays **[exploratory]** until that face lands. House rules bind; misses enter the ledger at equal prominence.